



Deep learning models for enhanced in-field maize leaf disease diagnosis

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ABSTRACT

Maize leaf diseases significantly threaten crop yields, and there is need for accurate, and accessible diagnostic tools. This research addresses this need by developing and evaluating deep learning (DL) and machine learning (ML) models for in-field classification and detection of four critical maize diseases: Maize Leaf Blight, Maize Lethal Necrosis, Maize Streak Virus, and Fall Armyworm damage. Utilizing field imagery captured via digital cameras and smartphones across Uganda, Tanzania, Ghana, and Namibia, we developed and compared custom Convolutional Neural Networks (CNNs), transfer learning (MobileNetV2, InceptionResNetV2), Vision Transformers (ViT), and classical ML models. For detection, a transformer-enhanced YOLOv10 architecture was implemented. Explainable AI (XAI) techniques (Grad-CAM, LIME) were incorporated to ensure model transparency. MobileNetV2 achieved the highest classification accuracy (97%), while the enhanced YOLOv10 reached a mean Average Precision (mAP) of 0.995 for object detection. The best models were integrated into a mobile application deployed on edge devices for real-time diagnosis by smallholder farmers in Uganda, with performance validated through field tests. This study demonstrates a powerful, interpretable, and field-deployable solution combining advanced DL, transformers, and XAI for effective maize health management.

1. Introduction

Maize is a globally grown cereal crop and ranks as the highest-produced grain worldwide. It is crucial in supporting food security; it is also used for livestock feed and energy production, especially in the context of a rising global population in the African context (Qian et al., 2022). In addition to its nutritional importance, maize is a key raw material in the manufacture of several industrial products.

Its widespread use as a staple food for humans and animals underscores its economic and nutritional value (Pacal, 2024). However, the sustainability of maize production is increasingly threatened by a range of diseases and pests that pose serious risks to global food security. These include Maize Leaf Blight (MLB), Maize Lethal Necrosis (MLN), Maize Streak Virus (MSV), and Fall Armyworm (FAW). MLB, caused by *Bipolaris maydis*, results in leaf lesions and yield losses due to reduced photosynthesis (Shah et al., 2024). MLN arises from Maize Chlorotic Mottle Virus (MCMV) and Potyviruses infection, often leading to plant death. MSV, transmitted by leafhoppers, causes stunted growth and severe yield reduction (Shah et al., 2024). Fall Army Worm (*Spodoptera frugiperda*) inflicts damage through defoliation and destruction of the plant's central whorl (Ajam et al., 2024).

Given the widespread consumption of maize and its importance in agriculture, it is crucial to improve methods for accurately detecting and classifying its diseases and pests. While various efforts have been made to utilize image-based methods for disease detection, many existing solutions face challenges such as scalability, poor performance on low-resource devices, and the inability to generalize across different disease types. This is particularly evident in Sub-Saharan Africa, where there is a need for the development of robust, lightweight, and accurate models that can be deployed effectively in rural field conditions. Additionally, timely and accurate disease management is essential; however, many farmers in these regions lack access to expert advisory services, which hampers their ability to respond effectively to crop threats (Setiawan et al., 2022b). Traditional diagnostic methods typically rely on expert visual inspections, which can be subjective, time-consuming, and prone to errors. And misdiagnosis may lead to the overuse of pesticides or inadequate treatment, ultimately resulting in yield loss or crop failure.

These challenges highlight the urgent need for low-cost, accurate, and readily available diagnostic tools to support sustainable farming, particularly in developing countries. This research addresses the

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challenge by leveraging computer vision (CV) and deep learning technologies for image classification and object detection of maize diseases and pest damage. By deploying high-performing models on edge devices such as mobile applications, smallholder farmers can quickly identify maize diseases early for management, hence combating further disease spread.

Several prior studies have explored AI-based solutions for maize disease detection. [Setiawan et al. \(2022a\)](#) compared machine learning (ML) and deep learning (DL) approaches across 62 studies, finding deep learning models more effective in handling complex and variable conditions due to their superior feature extraction capabilities. [Ahadian et al. \(2024\)](#) demonstrated the importance of model and loss function selection, achieving 97.15% accuracy with VGG16 and strong performance with EfficientNet on imbalanced data. [Pacal \(2024\)](#) introduced a lightweight multi-axis vision transformer (MaxViT) model that outperformed others in accuracy and speed, making it suitable for resource-limited environments. Researchers in [O'Halloran et al. \(2024\)](#) developed deep learning models for Maize Lethal Necrosis and Maize Streak Virus disease, a portion of the maize disease covered in this paper. Their research showed that the EfficientNet V2 model had the best accuracy results with a 99.9% in disease classification.

[Haque et al. \(2022\)](#) achieved 99.14% accuracy in identifying Maydis leaf blight using a GoogleNet-based deep CNN, emphasizing the model's ability to distinguish fine-grained disease features. [Amin et al. \(2022\)](#) developed a hybrid model combining EfficientNetB0 and DenseNet121, surpassing more complex architectures and highlighting the potential of hybrid approaches. [Richey et al. \(2020\)](#) developed a real-time mobile application using ResNet50 and transfer learning, providing immediate feedback and a scalable solution for early disease detection. [Kalyango and Ntanda \(2023\)](#) proposed an interpretable deep learning framework for diagnosing MSV using SHAP and LIME, offering transparency crucial for user trust in agricultural AI systems.

Despite this progress, key research gaps remain and addressing these gaps requires a multifaceted approach focused on improving dataset quality, enhancing model robustness and efficiency across a larger maize diseases that affect the African continent. The following are the main research gaps and contributions of this paper.

1. *Lack of dataset diversity and real-world representation:* Existing studies depend on limited datasets that do not capture all disease manifestations across different geographical and environmental contexts ([Ahadian et al., 2024](#)). Furthermore, many datasets are collected under controlled conditions, reducing the generalization of machine learning models to real-world scenarios ([Wang et al., 2024](#)). For example, images from PlanetVillage captured controlled environmental settings and did not capture real in-field maize settings ([Haque et al., 2022](#)). This study addresses this gap by utilizing diverse datasets collected from in-field agricultural settings across varied regions, crop diseases and environmental conditions in Africa, thereby improving model robustness and enhancing generalization.
2. *Limited optimization and comparative evaluation of deep learning approaches:* Although numerous studies have explored deep learning for maize disease detection, mainly through transfer learning, there is insufficient analysis of how different model architectures and optimization strategies perform under varying data conditions ([Amin et al., 2022](#); [Hu et al., 2024](#); [Madhavi et al., 2024](#); [Niyomwungere et al., 2022](#); [Rai & Pahuja, 2024](#)). This study fills that gap by comprehensively evaluating classical ML models, custom CNNs, transfer learning techniques, and Vision Transformers. It proposes optimization strategies using tools like Keras Tuner to identify the most effective maize leaf disease identification model configurations.
3. *Lack of Explainable AI (XAI) integration:* While a few studies have introduced explainable AI tools, their use remains limited ([Banadda et al., 2023](#); [Kalyango & Ntanda, 2023](#)). This

study incorporates advanced XAI methods like SHAP and LIME to improve transparency and model interpretability. By enabling smallholder farmers and agricultural experts to understand how decisions are made, this research enhances trust in AI models, allowing the adoption of AI models in practical settings.

4. *Gap between research prototypes and real-world deployment:* Many AI models have shown been tested in lab settings but remain untested in field settings ([Haque et al., 2022](#); [Mohanty et al., 2023](#); [Richey et al., 2020](#)). There is a gap in translating AI research into scalable, farmer-friendly tools. To address this, the study presents a mobile application that integrates the best-performing image classification model optimized for deployment on edge devices. The mobile application has been validated through field trials with agricultural experts and smallholder farmers, demonstrating its usability and potential for large-scale in-field adoption.
5. *Lack of linkage between AI development and agronomic expertise:* Prior research often lacks the integration of domain-specific knowledge, resulting in models that are impractical in the agricultural domain. This study discusses an approach that involves agricultural experts throughout development, from data collection and annotation to model evaluation and model testing. This interdisciplinary approach ensures that the model is technologically advanced and aligned with real-world agricultural settings.

The rest of the paper is organized as follows: Section 2 describes the methodology and detailed architecture of the proposed framework. Section 3 presents the detailed results, and discussion follows in Section 4. We conclude the paper in Section 5.

2. Methods

2.1. Approach

[Fig. 1](#) illustrates the methodology employed for in-field maize leaf disease diagnosis. The first step is data collection, cleaning, annotation, and preparation. After which, the datasets are used as input for two machine learning tasks for model training: image classification and object detection of maize diseases, utilizing datasets that capture diverse conditions across various geographical regions. The methodological workflow depicted in the approach also integrates XAI for model interpretability and edge deployment on mobile applications to facilitate robust in-field disease management and subsequent field evaluation. The methodology uses five distinct datasets: four datasets comprising images of healthy and diseased maize leaves collected from diverse African regions (Uganda, Tanzania, Namibia, and Ghana), and a fifth dataset containing a variety of random objects to constitute the 'unknown' class that was used during model training.

2.2. Datasets

This study utilized five distinct datasets, each representing diverse conditions of maize leaves, including both healthy and diseased image datasets. These datasets were collected from four African countries: Uganda, Tanzania, Namibia, and Ghana. The datasets capture different environmental conditions and disease manifestations. Each dataset comprised high-resolution images detailing specific conditions of maize leaves. Additionally, to ensure quality and consistency across datasets collected from these regions, standardized data collection protocols were employed. Field images were reviewed and validated by agricultural experts in each region to ensure accurate disease labeling. We also conducted cross-region checks to identify and correct inconsistencies in annotation or image quality, helping to harmonize the datasets and improve the generalizability of the models across different environmental and visual conditions.

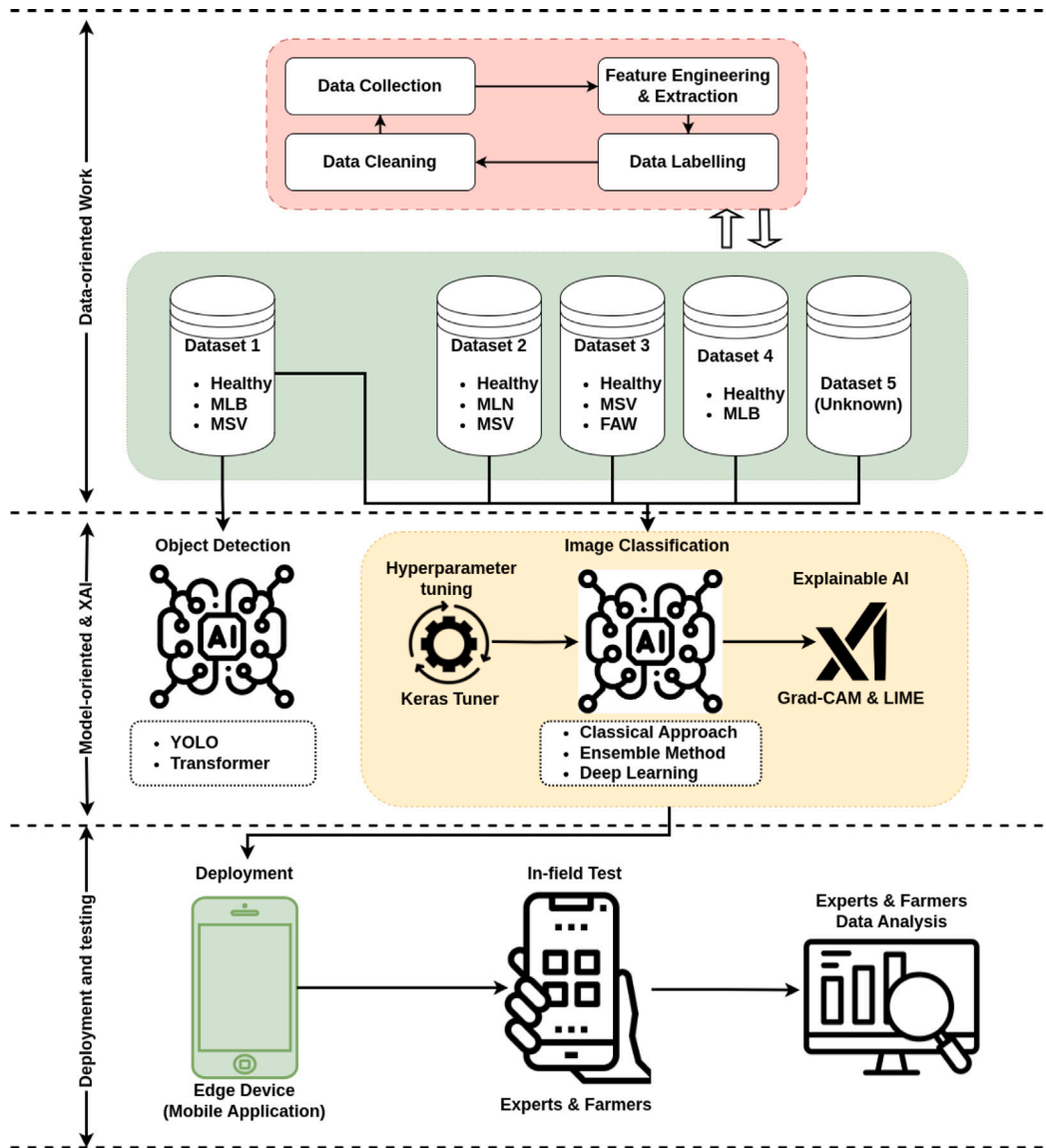


Fig. 1. The system architecture illustrates the relationships among various workflow processes and tasks. It begins with data-oriented work, which includes data collection, cleaning, extraction, and labeling. Next is the model-oriented flow, with classification and detection modeling, hyperparameter tuning, and explainable AI. Finally, deployment and testing involve model deployment, in-field testing, and data analysis.

- Dataset 1** was acquired by the Makerere AI Lab at Makerere University in Uganda contains images of healthy maize leaves and leaves exhibiting symptoms of Maize Leaf Blight (MLB) and Maize Streak Virus (MSV) (Babirye et al., 2022).
- Dataset 2** was acquired by the Nelson Mandela African Institution in Tanzania includes images of healthy maize leaves and those exhibiting symptoms of Maize Streak Virus (MSV) and Maize Lethal Necrosis (MLN) (Mduma et al., 2022).
- Dataset 3** was acquired by Namibia University in Namibia includes images of healthy maize leaves and those exhibiting symptoms of Maize Streak Virus (MSV) and damage by Fall Armyworm (FAW) (Blessing et al., 2022).
- Dataset 4** was acquired by KaraAgroAI in Ghana, includes images of healthy maize leaves and damage by Fall Armyworm (FAW) (Akogo et al., 2022).
- Dataset 5** or the 'unknown' class constitutes a diverse collection of randomly selected images from online image repositories. These images contain a variety of objects unrelated to maize leaf diseases and were added as an 'unknown' class during model training. This heterogeneity is crucial for developing a robust

model that accurately identifies and classifies images outside the defined maize leaf disease categories as belonging to the 'unknown' class.

Fig. 2 illustrates examples of the maize leaf diseases, highlighting key visual characteristics of each disease. The visual distinctions among these diseases are critical for developing accurate and effective models for disease identification and classification.

- Maize Streak Virus (MSV):** Characterized by distinct yellowish streaks parallel to leaf veins, disrupting photosynthesis and affecting plant growth and yield.
- Maize Lethal Necrosis (MLN):** Manifests as yellowing or chlorotic streaks from leaf tips downwards, potentially leading to uniform yellowing and necrosis; caused by multiple viruses and highly destructive.
- Maize Lethal Blight (MLB):** Characterized by elongated, light brown necrotic lesions, often with a yellow halo, that can spread across the leaf causing significant damage.

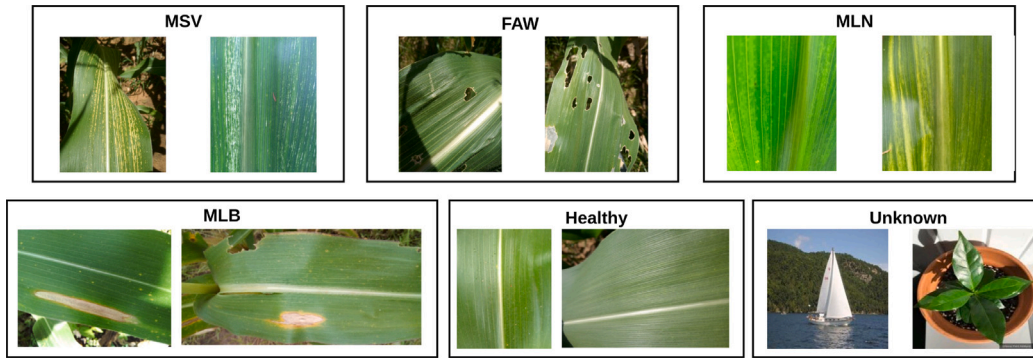


Fig. 2. Sample images representing the classes considered. In addition to the target classes of maize diseases: Maize Streak Virus (MSV), Maize Lethal Necrosis (MLN), Maize Leaf Blight (MLB), Fall Armyworm damage (FAW), healthy plants, and the ‘unknown’ class.

Table 1

Distribution of images across training, validation, and test sets for each class.

Class	Train	Validation	Test	Total
MLN	2786	796	398	3980
MSV	9515	2718	1361	13594
MLB	3695	1055	529	5279
FAW	935	267	135	1337
Healthy	9132	2609	1305	13046
‘Unknown’	934	200	201	1335

4. Fall Armyworm (FAW): Characterized by irregular holes scattered on leaves and feeding damage along leaf edges, representing physical destruction distinct from pathogen-caused diseases.
5. Healthy maize leaves: Exhibit a uniform deep green color with no discoloration, necrosis, or physical damage; serve as the baseline for disease detection.
6. Unknown Class: Includes non-maize images of unrelated objects or plants, critical for testing the model’s ability to filter irrelevant images and prevent model misclassification.

To preserve the natural diversity in field conditions, including lighting, background, and image quality, no explicit preprocessing was performed to standardize these variations. Instead, model-specific preprocessing pipelines, such as normalization and resizing, were applied during training to enhance robustness and help models generalize across different imaging conditions. This targeted approach was adopted to optimize the performance of each model architecture. Additionally, since this dataset was derived from real-world field conditions, it exhibited natural class imbalance across categories, with certain classes, such as FAW and MLN, being underrepresented. No artificial balancing methods (e.g., oversampling or synthetic data generation) were applied, as the goal was to maintain the integrity of naturally occurring distributions. However, stratified random sampling was employed during dataset partitioning to ensure proportional class representation across training, validation, and test sets. This approach aimed to prevent skewed model training while preserving the dataset’s ecological validity.

The images from the four maize leaf datasets (healthy and diseased) were combined with the ‘unknown’ class dataset to form the complete dataset. The dataset was divided into a 70% training set, a 20% validation set, and a 10% testing set using a stratified random sampling approach to maintain proportional class representation within each subset. The distribution of images across these subsets for each class is detailed in Table 1.

2.3. Image classification

Image classification distinguishes object categories using attributes via classical machine learning or advanced neural networks (Mishra

et al., 2023). Classical methods employ feature extraction (e.g., Speeded-Up Robust Features (SURF), Scale-Invariant Feature Transform (SIFT)), which play a key role in identifying distinct features crucial to maize leaf classification (Supriya & Khatoon Mohammed, 2024). Then, algorithms such as k-NN and Random Forest are used for feature classification. Deep learning, however, offers superior performance, particularly CNNs (e.g., VGG, ResNet, Inception, MobileNet, EfficientNet). Transfer learning enables a model to adapt pre-trained models to new tasks, which is beneficial for tasks with limited datasets. YOLO models are efficient for object detection and suitable for image classification on mobile phones. Evaluating model efficiency and using XAI for interpretability are crucial (Kamakshi & Krishnan, 2023).

2.3.1. Classical approach

1. Feature Extraction using SIFT:

The Scale-Invariant Feature Transform (SIFT) algorithm was chosen for robust local feature extraction due to its invariance to scale, rotation, and affine transformations. The model generates a 128-dimensional descriptor vector for each key-point, making SIFT an excellent choice for feature extraction in the maize leaf datasets. The models were implemented using OpenCV with optimized parameters. SIFT effectively identified key points in healthy, MSV-affected, and MLN-affected maize leaves and non-maize objects, as shown in Fig. 3. As observed, SIFT captured essential differentiating features, evidenced by the highlighted streaks in MSV and lesions in MLN, with fewer key points detected in the smoother texture of healthy leaves. This demonstrates SIFT’s versatility in identifying informative regions crucial for distinguishing between different classes.

2. **Classification with KNN:** After feature extraction using SIFT, the resulting descriptor vectors were classified using the K-Nearest Neighbors (KNN) algorithm. This non-parametric, instance-based learning method classifies a sample based on the majority class of its k nearest neighbors within the feature space, where k was optimized via cross-validation and Euclidean distance was selected as the distance metric (Cunningham & Delany, 2021). The selection of KNN was motivated by its simplicity and effectiveness in classifying the relatively low-dimensional feature space produced by the SIFT descriptors. The Euclidean distance between two points x and y in an n -dimensional feature space is calculated using the following formula:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (1)$$

where $d(x, y)$ is the Euclidean distance between two points x and y , and n is the number of features.

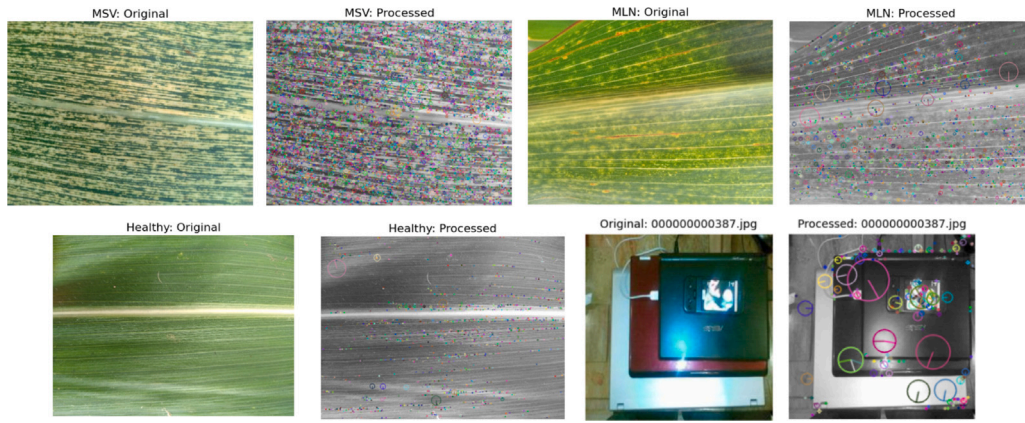


Fig. 3. Sample images showing the SIFT features extracted for training the classification model illustrate the original images with the corresponding SIFT features extracted.

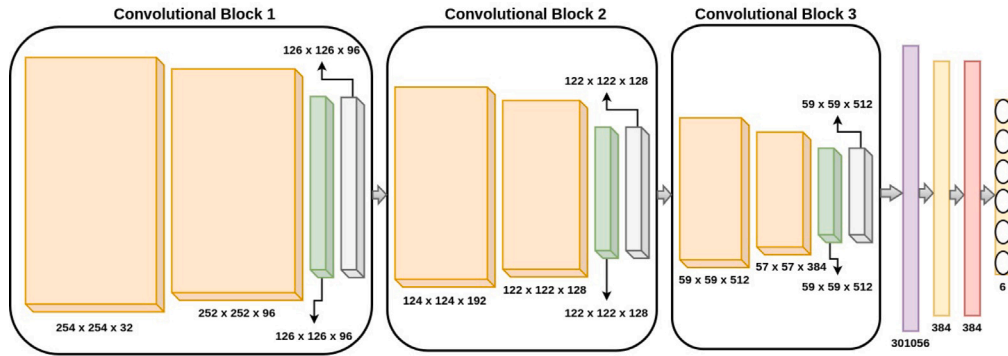


Fig. 4. Architecture of the convolutional neural network used as the custom model for classification. It illustrates three main convolutional blocks followed by fully connected layers that reduce dimensionality and output classification probabilities of the six considered classes.

Table 2

Hyperparameter tuning results for the custom CNN model.

Hyperparameter	Ranges	Best value
Conv1 filters	32, 64, 96, 128	32
Conv2 filters	32, 64, 96, 128	96
Conv3 filters	64, 128, 192, 256	192
Conv4 filters	64, 128, 192, 256	128
Conv5 filters	128, 256, 384, 512	512
Conv6 filters	128, 256, 384, 512	384
Dense units	128 to 512 (step of 128)	384
L2 regularization	0.001, 0.0001, 0.00001	0.001
Dropout rate	0.3 to 0.7 (step of 0.1)	0.4
Learning rate	1e-2, 1e-3, 1e-4	0.0001

Table 3

Fixed hyperparameters for ResNet50 and EfficientNetB0 models.

Model	Hyperparameters	Values
ResNet50	Dense units	1024
	Dropout rate	0.2
	Learning rate	0.0001
EfficientNetB0	Dense units	1024
	Dropout rate	0.2
	Learning rate	0.0001

2.3.2. Deep learning

- Custom CNN Architecture:** The second method involved the development of a custom Convolutional Neural Network (CNN) for maize leaf classification, as shown in Fig. 4. This model was developed from scratch with an architecture comprising six sequential convolutional layers, each likely followed by batch

normalization and a ReLU activation function. This was succeeded by a fully connected layer with a variable number of units, incorporating batch normalization and a ReLU activation. Dropout layers were included to enhance generalization, and the output layer used a softmax activation function to produce class probabilities.

The custom CNN architecture comprises six convolutional layers and a fully connected layer with varying dense units. To optimize these units, we employed Keras Tuner's RandomSearch technique. This method systematically explores different hyperparameter combinations within predefined ranges to maximize validation accuracy (Hedhoud et al., 2023). Hyperparameter tuning was performed to identify the optimal values for several parameters, including the number of filters in each convolutional layer, the number of units in the dense layer, the strength of L2 regularization, the dropout rate, and the learning rate. The best-performing hyperparameter configuration for the custom CNN model is summarized in Table 2.

- Transfer Learning:** The third approach in this study involved using pretrained models such as ResNet50, EfficientNetB0, MobileNetV2, Xception, InceptionV3, and InceptionResNetV2 through transfer learning. These models were initially trained on the ImageNet dataset and then fine-tuned on the maize leaf datasets used in this study. The selection of these models was based on their proven performance in various computer vision tasks, providing a balance between depth, parameter efficiency, and representational power. Lightweight models, suitable for edge deployment, were included, such as MobileNetV2 and EfficientNetB0, alongside deeper architectures known for their high precision, including Xception and InceptionResNetV2. The fine-tuning process included freezing the first layers of the models and retraining the final layers to adapt to the specific

Table 4
Hyperparameter tuning results for various models.

Model	Hyperparameters	Ranges	Best hyperparameters
MobileNetV2	Dense units	128 to 512 (step of 128)	384
	L2 regularization	0.001, 0.0001, 0.00001	1e-05
	Dropout rate	0.3 to 0.7 (step of 0.1)	0.6
	Learning rate	1e-2, 1e-3, 1e-4	0.0001
Xception	Dense units	128 to 512 (step of 128)	384
	L2 regularization	0.001, 0.0001, 0.00001	1e-05
	Dropout rate	0.3 to 0.7 (step of 0.1)	0.5
	Learning rate	1e-2, 1e-3, 1e-4	0.0001
InceptionV3	Dense units	128 to 512 (step of 128)	512
	L2 regularization	0.001, 0.0001, 0.00001	0.0001
	Dropout rate	0.3 to 0.7 (step of 0.1)	0.4
	Learning rate	1e-2, 1e-3, 1e-4	0.0001
InceptionResNetV2	Dense units	128 to 512 (step of 128)	384
	L2 regularization	0.001, 0.0001, 0.00001	1e-05
	Dropout rate	0.3 to 0.7 (step of 0.1)	0.3
	Learning rate	1e-2, 1e-3, 1e-4	0.0001

task of classifying maize diseases. This approach greatly reduced training time while maintaining high accuracy by using the detailed feature representations learned by these models on large-scale datasets.

The pretrained models were approached using fixed hyperparameters and optimized hyperparameters. The fixed hyperparameters, such as the number of dense units, dropout rate, and learning rate, were carefully chosen for the ResNet50 and EfficientNetB0 models, as summarized in Table 3. Keras Tuner's RandomSearch technique optimized hyperparameters for other models like MobileNetV2, Xception, InceptionV3, and InceptionResNetV2. The best-performing hyperparameters identified through this process are detailed in Table 4.

The models were trained using the Adam optimizer for model training and optimization, with an initial learning rate of 0.001. Cross-entropy loss was used as the loss function, and early stopping was implemented to prevent overfitting. The training was conducted on a GPU-enabled environment to accelerate the process, and extensive hyperparameter tuning was performed using grid search to find the optimal combination of batch size, learning rate, and dropout rate. The cross-entropy loss function is defined as:

$$L_{\text{cross-entropy}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (2)$$

where $L_{\text{cross-entropy}}$ is the cross-entropy loss, N is the number of samples, C is the number of classes, $y_{i,c}$ is the true label, and $\hat{y}_{i,c}$ is the predicted probability for class c .

- 3. Vision Transformer (ViT):** The Vision Transformer (ViT) architecture, specifically the “vit_B_16_weight” variant, was utilized for this approach. ViT processes images by dividing them into 16×16 pixel patches, which are linearly embedded, augmented with positional embeddings, and processed through a standard transformer encoder. The model was initialized with pre-trained weights from a substantial image dataset and subsequently fine-tuned on the maize leaf datasets using the AdamW optimizer, selected for its efficacy with large-scale models like transformers. Hyperparameter optimization focused on the learning rate, batch size, and weight decay. The ViT-B/16 model is able to capture global contextual information within the maize leaf images. The specific model training hyperparameters are detailed in Table 5.

Hyperparameter tuning was conducted explicitly for the CNN classification models to optimize parameters. For other architectures, default or empirically suggested configurations were utilized due to their architectural complexity and pre-established training regimes.

Table 5
Hyperparameters used for training the Vision Transformer (ViT) model.

Hyperparameter	Description	Value
Patch size	Size of image patches	16×16
Optimizer	Optimization algorithm	AdamW
Learning rate schedule	Learning rate decay strategy	Cosine schedule

2.3.3. Ensemble method

Model Selection: This study employed an ensemble method that leveraged the complementary strengths of two distinct deep learning architectures: a small convolutional neural network (convnet_small_in22k) (Liu et al., 2022) and a Vision Transformer (vit_small_patch16224). The convnet_small_in22k was selected for its efficiency and proven ability to effectively capture spatial hierarchies within image data, a crucial aspect for analyzing leaf patterns. Conversely, the vit_small_patch16224 was incorporated to model long-range dependencies across the image. By treating images as sequences of patches, ViT excels at capturing global contextual information. Both models were pre-trained on large image datasets, enabling more effective fine-tuning on maize leaf datasets.

Training Procedure: The ensemble model was trained using the FastAI library, leveraging advanced data augmentation and mixed-precision training. An optimal learning rate was determined using FastAI's learning rate finder, ensuring efficient convergence during fine-tuning. Gradual accumulation was employed to manage memory constraints while utilizing larger batch sizes, thereby stabilizing the training process. The learning rate finder identified two potential learning rates, as depicted in Fig. 5: 0.0017, corresponding to the valley point, and 0.0030, corresponding to the slide point. These values were then used to guide the model training, optimizing the model convergence process.

Ensemble Strategy: The classification outputs from the ConvNeXt-Small and Vision Transformer Small models were combined using a weighted voting scheme. The final prediction was determined by the weighted sum of the class probabilities predicted by each model. The weights assigned to each model were proportional to their respective validation F1-scores achieved during fine-tuning, ensuring that the ensemble leveraged the strengths of the better-performing model(s). This strategy aimed to improve overall classification accuracy by reducing the impact of individual model biases and errors. The weighted voting scheme can be expressed as:

$$P_{\text{ensemble}}(c) = w_1 \cdot P_{\text{convnet}}(c) + w_2 \cdot P_{\text{vit}}(c) \quad (3)$$

where $P_{\text{ensemble}}(c)$ is the ensemble probability for class c , $P_{\text{convnet}}(c)$ and $P_{\text{vit}}(c)$ are the probabilities assigned to class c by the ConvNet and ViT models respectively, and w_1 and w_2 are the corresponding weights.

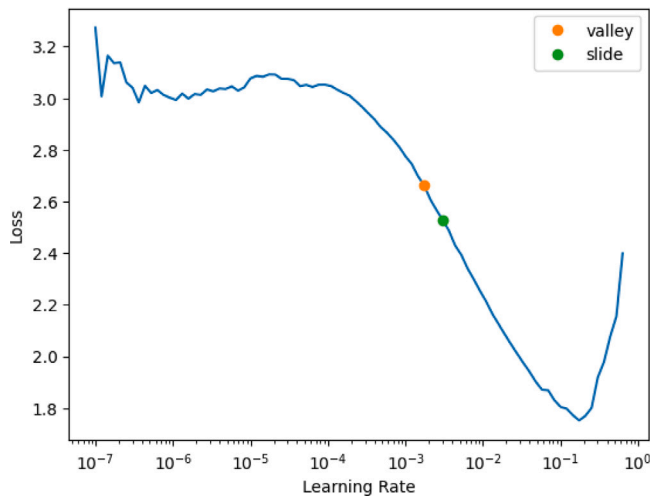


Fig. 5. Learning Rate Finder Curve. The plot shows the loss as a function of the learning rate.

Prediction Aggregation: During inference, predictions from the ConvNet and ViT models were aggregated using a weighted voting strategy. Each model assigned a confidence score to its predicted class, and these scores were combined based on predefined weights that reflected each model's validation accuracy and class-specific performance. In cases of conflicting predictions, the final decision favored the class with the highest weighted confidence, ensuring robustness while leveraging the complementary strengths of both architectures.

Evaluation and Metrics: The ensemble model's performance was evaluated on a held-out test set using standard classification metrics, including F1-score, precision, recall, and accuracy. A confusion matrix was generated to provide a detailed visualization of the model's classification performance across different maize leaf conditions and the 'unknown' class, highlighting any potential areas of confusion between specific categories.

2.3.4. Model explainability

Explainable AI techniques such as Local Interpretable Model-Agnostic Explanations (LIME) (Ribeiro et al., 2016) and Gradient-weighted Class Activation Mapping (GradCAM) (Selvaraju et al., 2017) were integrated to provide insights into the decision-making process of the deep learning models. LIME generated interpretable models for individual predictions by perturbing the input and observing the prediction changes. GradCAM was applied to visualize the regions of the input image that contributed most to the model's decision, highlighting the areas the model focused on. The outputs from LIME and GradCAM were integrated into the classification pipeline to enhance the interpretability of the model results. These explanations were particularly useful in validating the model's predictions by ensuring that the model focused on biologically relevant features such as lesion patterns and color changes in maize leaves.

2.4. Object detection

Unlike image classification, object detection categorizes multiple objects within an image, providing precise location coordinates. This capability is essential for maize leaf analysis, enabling the isolation of individual leaves and lesions on the leaves. Prominent object detection models include YOLO, valued for its real-time efficiency, and Vision Transformers like DETR, which streamline detection by framing it as a direct set prediction task (Han et al., 2023).

YOLO Models: This study utilized the YOLO (You Only Look Once) family of models, known for efficient real-time object detection, to

identify various conditions of maize leaves. We specifically developed and fine-tuned successive versions of the model: YOLOv8, YOLOv9, and YOLOv10 (Alif & Hussain, 2024). YOLOv8 improved upon previous YOLO models by introducing a more compact architecture, enhancing speed and accuracy through reduced layers, pre-training on large datasets, and refined handling of small objects and non-maximum suppression (NMS) to minimize false positives. YOLOv9 further enhanced detection accuracy with deeper convolutional layers, advanced anchor box scaling, and an adaptive learning rate scheduler, significantly improving the detection of multiple disease symptoms on a single leaf. Finally, YOLOv10 integrated transformer-based components and a sophisticated feature pyramid network (FPN) to leverage global image context and improve multi-scale detection, outperforming prior versions in both speed and accuracy on maize leaf datasets, making it highly suitable for real-time applications (Bochkovskiy et al., 2020).

DETR Transformer: The Detection Transformer (DETR) model is an object detection model that replaced the convolutional approaches with a transformer-based architecture that directly predicts object sets (Carion et al., 2020). By leveraging a transformer encoder-decoder, DETR captures global relationships and eliminates hand-crafted components like anchor boxes and non-maximum suppression, thus simplifying training. Fig. 6 shows examples of sample images with bounding boxes allocated as labels for Maize Lethal Blight and Maize Streak Virus diseases.

2.5. Mobile application development and in-field test

We developed a mobile application, *MaizeNet* using the React Native framework. This was important to enable us deploy the optimal classification model for real-time maize leaf analysis on Android and iOS platforms. The application features a user-friendly interface enabling image capture and gallery upload, with on-device inference model optimized for resource-constrained devices via model quantization and batching. This design ensures low-latency, real-time user feedback, including an expert feedback option for model retraining. Rigorous testing validated the application's accuracy and performance on mobile platforms.

Finally, we carried out a field test of the *MaizeNet* application. This was done together with agricultural experts and farmers across diverse maize fields in Uganda. The maize images were collected following a standardized protocol where maize images were taken from 25–40 cm to capture detailed symptoms, focusing on diseased leaf portions under varying light conditions. The emphasis was on detailed symptom capture and user-guided image cropping within the app. The agricultural experts validated the model diagnoses and provided feedback. All data was uploaded to Firebase, ensuring a comprehensive dataset for model refinement and improved field accuracy.

3. Results

In this section, we present the experimental results of the classification and detection model for maize disease diagnosis.

3.1. Classical machine learning models

In the first method, we used the traditional method using SIFT for feature extraction, followed by classification with KNN, which showed good performance. As shown in Table 6, the model achieved a training accuracy of 92% and a test accuracy of 89%, with precision, recall, and F1-score at 88%. These results suggest that classical machine learning methods can establish a good baseline for maize disease diagnosis. However, the classical machine learning models do may not achieve the same level of performance as more advanced deep learning models, mainly when dealing with diverse and complex datasets.

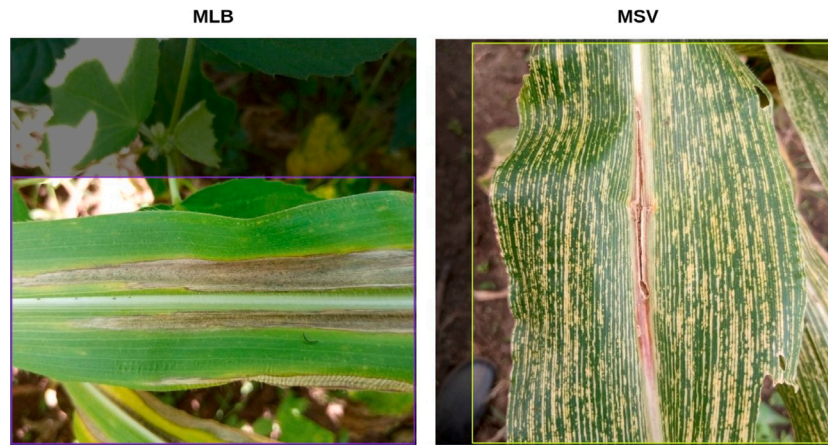


Fig. 6. Two sample images illustrate how bounding boxes were allocated as labels for Maize Lethal Blight and Maize Streak Virus diseases. These bounding boxes were drawn around the leaf or the portion of the image affected by a specific maize disease.

Table 6

Model testing results for the classical machine learning models.

Accuracy	Precision	Recall	F1-Score
0.89	0.88	0.88	0.88

Table 7

Evaluation metrics for the custom CNN model.

Evaluation metric	Value
Train loss	0.30
Train accuracy (%)	0.99
Validation loss	0.64
Validation accuracy (%)	0.90
Test accuracy (%)	0.91
Test precision (%)	0.92
Test recall (%)	0.91
Test F1-Score (%)	0.90

3.2. Deep learning models

Custom CNN Architecture: We developed a custom CNN model that showed strong performance in classifying maize leaf diseases better than the classical machine learning approaches. The CNN models achieved a training accuracy of 99% and a test accuracy of 91%. The precision was 92%, and recall was 91%, as detailed in Table 7. The F1-score was 90%, indicating a balanced performance between precision and recall. However, as shown in Fig. 7, the training and validation accuracy and loss over epochs indicate slight overfitting, with the training accuracy significantly higher than the test accuracy indicating slight overfitting and suggesting potential limitations in generalizing to unseen maize leaf images.

Fig. 8 shows a confusion matrix for the custom CNN model. The analysis shows a strong classification performance for Healthy and MSV classes. However, the model exhibits a tendency to misclassify MLB, confusing it with both MSV and healthy, suggesting potential feature overlap or class imbalance.

3.2.1. Transfer learning

The third approach used transfer learning with pre-trained models, specifically MobileNetV2 and InceptionResNetV2, which demonstrated strong performance across training, validation, and testing phases. MobileNetV2 achieved a training loss of 0.07 and a validation loss of 0.11, coupled with training and validation accuracies of 97% and 96%, respectively as shown in Table 8. The performance metrics show the effectiveness of the learning and generalization capabilities of the models. InceptionResNetV2 also exhibited robust performance, with a training accuracy of 97%, a validation accuracy of 95%, and low

Table 8

Training and Validation metrics for the transfer learning models.

Model	Train loss	Train accuracy %	Val loss	Val accuracy%
EfficientNetB0	1.46	0.77	1.37	0.81
ResNet50	0.66	0.95	0.62	0.92
MobileNetV2	0.07	0.97	0.11	0.96
Xception	0.06	0.95	0.15	0.94
InceptionV3	0.09	0.94	0.21	0.93
InceptionResNetV2	0.07	0.97	0.12	0.95

Table 9

Test data metrics for the transfer learning models.

Model	Precision %	Recall %	F1-Score %	Accuracy %
EfficientNetB0	0.40	0.45	0.35	0.35
ResNet50	0.68	0.69	0.68	0.69
MobileNetV2	0.97	0.97	0.97	0.97
Xception	0.95	0.95	0.95	0.95
InceptionV3	0.94	0.94	0.94	0.94
InceptionResNetV2	0.96	0.96	0.96	0.96

training and validation losses of 0.07 and 0.12. Notably, both models maintained high performance on the test data as shown in Table 9, with MobileNetV2 achieving 97% and InceptionResNetV2 achieving 96% across accuracy, precision, recall, and F1-score. This shows that the models are able to capture the necessary features to differentiate between the disease classes in the maize leaf disease dataset.

The training and validation performance of the MobileNetV2 model, as shown in the accuracy and loss graphs 9, showcases the model's effective learning capabilities. As shown in the left graph, the training accuracy consistently increased, approaching 98% by the 20th epoch, while the validation accuracy steadily climbed to around 96%; this indicates that the model not only learned the features of the maize leaf disease dataset but also generalized well to the validation data. The right graph, showing the loss, further supports this observation; training and validation losses decreased significantly and converged to low values, suggesting that the model's predictions became increasingly accurate as training progressed.

The confusion matrix shown in Fig. 10 provides a detailed view of MobileNetV2's classification performance across different classes. The model accurately classified the Healthy and MSV classes, with 1274 and 1325 correctly classified samples. There were some minor misclassifications, primarily between classes with similar symptoms, such as MLB and MLN, where some instances were incorrectly classified into the other category. Despite these few errors, the overall accuracy remained high, with MobileNetV2 successfully classifying various disease types, healthy samples and 'unknown' class.

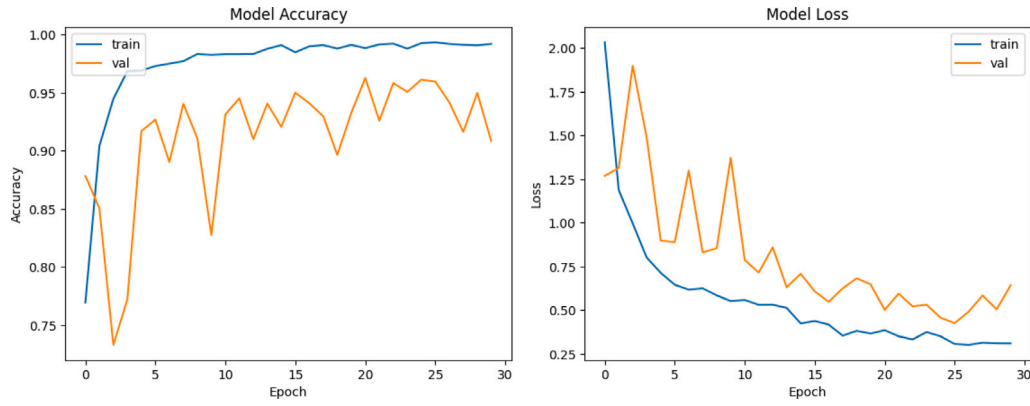


Fig. 7. Train and validation accuracy and loss for the custom Convolutional Neural Network model.

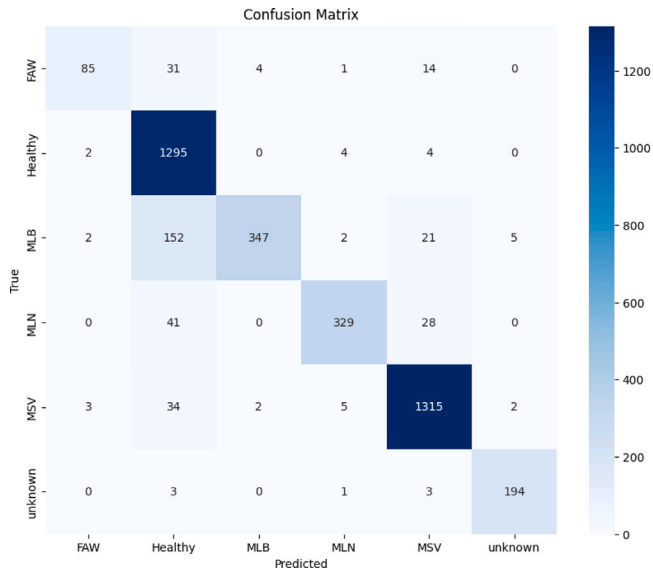


Fig. 8. The confusion matrix of the CNN shows classification performance across 6 maize classes. The model demonstrates high accuracy, with most true labels matching the predicted labels.

Table 10
Performance metrics for the Vision Transformer (ViT) model.

Metric	Value	Dataset
Train loss	0.02	Training
Train accuracy	0.99	Training
Validation loss	0.08	Validation
Validation accuracy	0.98	Validation
Test precision	0.97	Test
Test recall	0.98	Test
Test F1-Score	0.97	Test
Test accuracy	0.98	Test

3.2.2. Vision transformer models

The Vision Transformer model showed promising results for disease classification. It achieved a training accuracy of 99% and a test accuracy of 98%, as shown in Table 10. The model's precision and recall were 97% and 98%, respectively, with a high F1 score of 97%. The high accuracy and precision metrics indicate that the ViT model was able to effectively capture the features of maize diseases.

Table 11
Ensemble model performance metrics.

Metric	Training	Validation	Test
Loss	0.0027	0.0138	–
Accuracy	0.9982	–	0.9982
Precision	–	–	0.9719
F1-Score (Micro)	–	–	0.9982
F1-Score (Macro)	–	–	0.9723
F1-Score (Weighted)	–	–	0.9973
Error rate	–	–	0.0018

3.3. Ensemble model

The ensemble approach combined the outputs of the ConvNet and ViT and demonstrated strong performance as well across all metrics. The ensemble model achieved a training accuracy of 99.82% and a test F1-score of 99.72%, with an error rate of just 0.18%, as shown in Table 11. This combination strategy effectively reduced variance and improved prediction reliability, highlighting the benefits of using ensemble approaches in complex image classification tasks.

3.4. Explainable AI

Grad-CAM Analysis: The Grad-CAM visualizations as shown in Fig. 11 show that the model focused on disease-specific features for classification. For instance, FAW classifications were driven by lesion and hole patterns, while MSV predictions correlated with striped discolorations on the maize leaves. In the 'Unknown' class, the model highlighted non-typical disease areas. These visualizations validate the model's reliance on relevant visual cues, enhancing confidence in its predictions. This ability to visualize where the model is "looking" offers confidence in its predictions, as it shows that MobileNetV2 model based its classifications on relevant and accurate features.

LIME Analysis: The LIME visualizations as shown in Fig. 12 provided pixel-level explanations of the model's predictions. For MLB, LIME emphasized necrotic and yellowing areas, validating the model's focus on key disease symptoms. LIME also illustrated the model's ability to differentiate subtle disease variations, like MLN versus MLB. This ability to discern subtle differences and focus on disease-specific features, as demonstrated by LIME, highlights MobileNetV2's strength in handling complex classification tasks. Notably, for the 'Unknown' class, LIME highlighted non-disease-specific regions, demonstrating the model's capacity to identify novel or unclassified conditions, crucial for real-world applications.

3.5. Object detection

For the object detection models for maize disease detection, YOLOv10 model outperformed previous model versions (YOLOv8 and

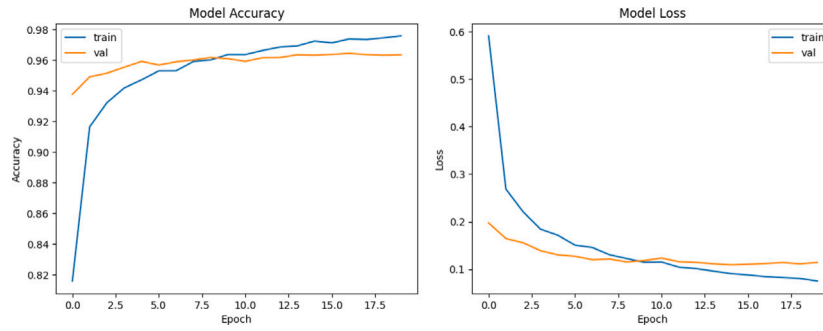


Fig. 9. Training and validation accuracy and loss for the best transfer learning model (MobileNetV2). The training generalized very well until early stopping.

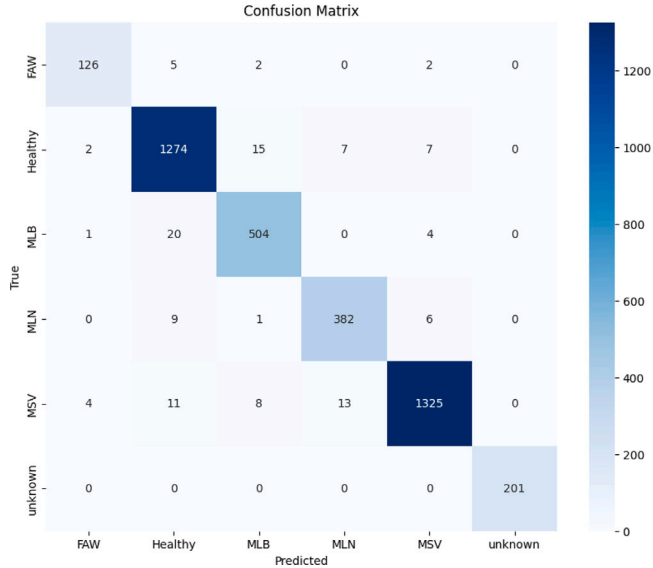


Fig. 10. The confusion matrix for the MobileNetB0 shows classification performance across 6 classes. The model demonstrates better performance than the custom CNN.

Table 12

YOLO Model Performance (mAP@0.5) on maize leaf datasets.

Metric/Class	YOLOv8	YOLOv9	YOLOv10
All classes (mAP@0.5)	0.973	0.994	0.995
MLB	0.993	0.994	0.994
MSV	0.994	0.994	0.994

YOLOv9). As shown in Table 12, YOLOv10 achieved an mAP of 0.995, demonstrating consistent performance across different disease classes. While YOLOv8 and YOLOv9 also performed well, YOLOv9 had a minor difference of only 0.001 compared to YOLOv10, and YOLOv8 was slightly less with 0.973.

The Detection Transformer (DETR) model showed a high performance for comparison, with an average precision (AP) of 0.873 at an IoU threshold of 0.50 to 0.95, as detailed in Table 13. DETR's unique approach to object detection, leveraging transformers, provided enhanced accuracy, particularly in detecting subtle and overlapping disease symptoms.

Fig. 13 illustrates the training metrics of YOLOv10, demonstrating a rapid initial reduction in box regression and classification losses, followed by stable convergence. This pattern indicates efficient feature learning for maize disease detection. The consistent decline in distribution focal loss (DFL) signifies optimized bounding box localization, crucial for precise detection. Furthermore, recall and mAP metrics stabilized near 1.0, highlighting the model's robust performance during training and validation.

Table 13

DETR object detection performance metrics.

Metric	Area	Value
AP @ IoU 0.50:0.95	All	0.873
AP @ IoU 0.50	All	0.981
AP @ IoU 0.75	All	0.891
AP @ IoU 0.50:0.95	Large	0.873
AR @ IoU 0.50:0.95	All	0.914
AR @ IoU 0.50:0.95	Large	0.914

The recall-confidence curve as shown in Fig. 14 shows YOLOv10's effectiveness in detecting maize leaf diseases. The model shows near-perfect recall across varying confidence thresholds. This high recall and precision confirm the model's ability to accurately identify disease instances in diverse scenarios. The minimal performance variance across MLB, MSV, and combined classes suggests consistent detection efficacy across disease types.

3.6. Qualitative evaluation and inference

Fig. 15 show the classification results of testing the MobileNetV2 model. The model is able to accurately identify diseases such as FAW, MLN, and MSV, with confidence levels close to or at 1.00. This indicates the effectiveness of MobileNetV2 in learning and distinguishing between different disease patterns. Additionally, the model performed well in classifying unknown instances or images, correctly identifying images that do not belong to predefined disease categories or healthy images of maize. The high consistency in confidence scores across different classes shows MobileNetV2's reliability and precision as a powerful model for deployment on edge devices to support smallholder farmers in disease diagnosis and management.

Fig. 16 shows the YOLOv10 model is able to effectively detect maize diseases. YOLOv10 effectively identified MLB and MSV, with high confidence scores, as indicated by the bounding boxes around the affected areas. The confidence scores, ranging from 0.88 to 0.94, reflect the model's strong ability to differentiate disease symptoms from healthy leaf regions.

3.7. Field testing and evaluation of the MaizeNet mobile application

To validate the practical application of the trained MobileNetV2 maize leaf disease detection model, a mobile application (*MaizeNet*) was developed for real-time disease classification. The application was evaluated using a test set of maize leaf images, non-maize images, and subsequently, in field tests with smallholder farmers and agricultural experts. The MaizeNet application, illustrated in Fig. 17, allows users to capture or upload maize leaf images for disease classification. The interface provides prediction results with confidence scores, indicating the likelihood of each disease. An expert input feature was implemented, enabling experts to provide labels and comments, enhancing diagnostic accuracy and facilitating data collection.

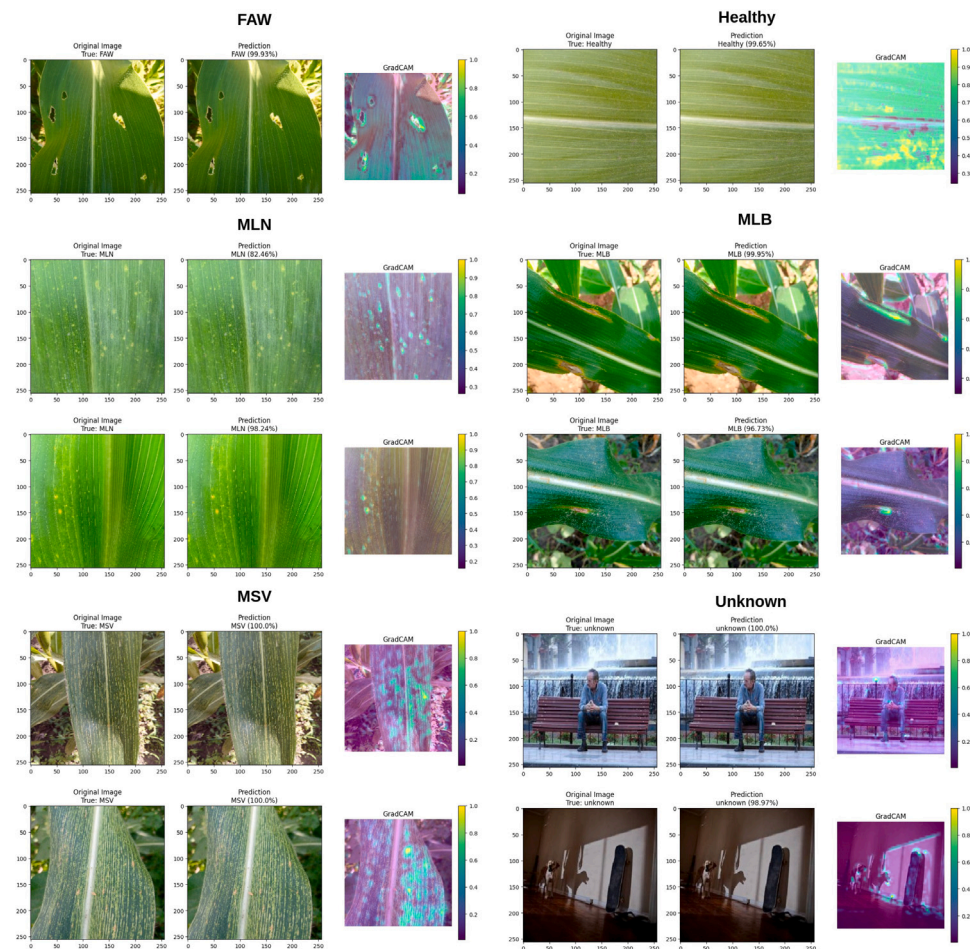


Fig. 11. Grad-CAM visualizations for interpreting model predictions from MobileNetV2. The columns show the original image, the predicted class with confidence score, and the corresponding Grad-CAM heatmap for each condition. The conditions include: FAW, MLN, MLB, MSV, and healthy leaves, with two examples for each category. Unknown images are also included for robustness testing. The heatmaps highlight the areas the model focused on for its predictions.

1. The application allows users to make predictions for maize images, which is particularly helpful for farmers who want to utilize the application without needing additional information.
2. For expert users, the application includes an option to provide their perspective on the predicted classes to override the model's predictions and the ability to add comments. This expert data collected after the application was used was later analyzed to assess the application's comprehensive performance.

The application workflow comprises the following steps:

1. **Image Acquisition:** The application's initial interface as shown in Fig. 17, top-left offers two image input modalities: real-time capture via the device's camera or selection from the local image gallery. This dual functionality supports immediate field analysis and retrospective analysis of stored maize images.
2. **Model Inference Initiation:** Upon image selection, the user proceeds to the classification interface (Fig. 17, top-middle) and activates the "Classify" function. This action triggers the embedded MobileNetV2 model to perform inference, generating a probability distribution across predefined maize disease classes.
3. **Prediction Results:** The inference results are presented on the third screen (Fig. 17, top-right and bottom-left), showing each disease class with its corresponding probability score. The class with the highest probability is selected as the primary predicted disease. This output allows users to assess the model's confidence and understand the relative likelihood of each disease.

4. **Expert Validation and Annotation:** For expert users, an additional validation layer is provided (Fig. 17, bottom row). Experts can either confirm the model's prediction or override it by selecting an alternative class from the "Expert's" dropdown menu. Furthermore, they can add textual annotations based on their visual assessment, providing contextual information and refining the diagnostic information about the image.
5. **Data Submission:** After expert validation and annotation, the "Submit" button (Fig. 17, bottom row) finalizes the data entry. This action transmits the data and resets the application interface to its initial state, preparing it for subsequent image analysis.

Field tests were conducted to evaluate the application's performance under real-world conditions. Fig. 18 shows the application used by a maize expert and a smallholder farmer.

The field test data was analyzed to assess the application's diagnostic accuracy and usability. The performance of the MaizeNet application during field testing was evaluated using a confusion matrix as shown in Fig. 21, which compares expert-assigned "Actual" labels with the model's "Predicted" labels for each maize disease class. This matrix provides a detailed overview of the model's classification accuracy across the tested disease spectrum.

The model accurately predicted most disease classes, as evidenced by analyzing 150 field-collected images. However, the Maize Lethal Necrosis (MLN) prediction posed a significant challenge. The rarity of MLN in the field, coupled with stringent containment protocols

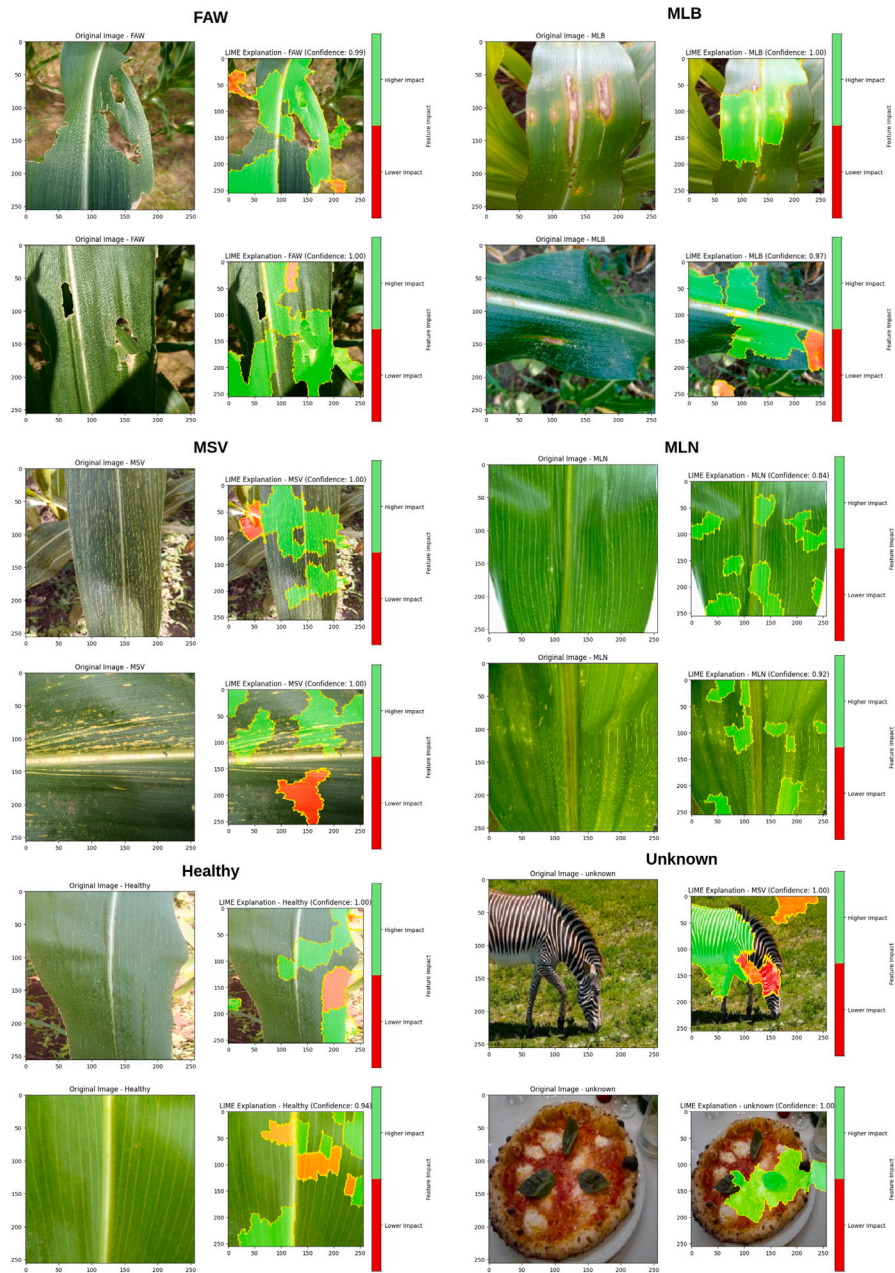


Fig. 12. LIME visualizations for interpreting model predictions from MobileNetV2. The columns show the original images and LIME explanation maps with confidence scores for different conditions. The green regions indicate areas of higher impact on the prediction, while red areas show lower impact. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

involving the destruction of infected fields, resulted in a limited dataset of only 24 MLN images. This scarcity of MLN samples likely contributed to the model's reduced performance in this class (see Fig. 19).

The figure compares actual and predicted counts for each maize disease class (excluding “Unknown”), with confidence intervals highlighting the model's reliability across categories. The classes displayed include FAW, Healthy, MLB, MLN, and MSV. Each point represents the count of images within each class, with blue circles indicating the expert-labeled (actual) data and red crosses showing the model's predictions. For most classes, such as FAW, Healthy, MLB, and MSV, the predicted values closely align with the actual values, showing minimal deviation and high model accuracy. The narrow confidence interval further supports the model's stability and consistent performance across these classes. However, a notable deviation appears with the MLN class, where the lower actual count reflects the challenges encountered in

field data collection due to the scarcity of MLN cases. Although the model's predictions for MLN are close to the actual count, the confidence interval is broader, indicating some variability in performance likely due to the limited sample size.

Furthermore, the ‘Unknown’ class was initially populated with 30 randomly selected images from non-target categories. An unexpected finding emerged when the expert recommended incorporating a new disease known as *maize rust* into the test set. The model tended to misclassify maize rust images as MLN, with 222 rust images being initially predicted as MLN. Consequently, these images were re-labeled as ‘Unknown’ by the agricultural experts. This misclassification highlighted a critical limitation of the model: its susceptibility to misinterpreting diseases outside its primary training set. This observation underscored the necessity for expanding the training dataset to include a wider range of maize diseases, thereby enhancing the model's robustness and

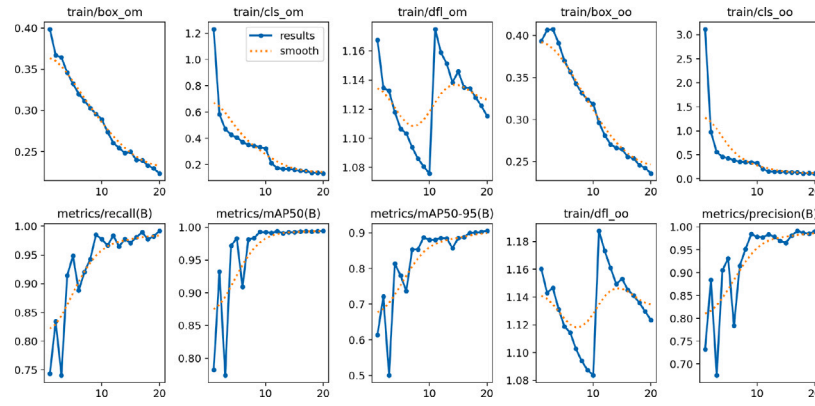


Fig. 13. Training and evaluation metrics for the YOLOv10 model. The top row shows training loss curves for bounding box regression (train/box), classification (train/cls), and distribution focal loss (train/df_l) across the objectness and no-objectness categories (_om and _oo). The solid blue lines represent raw results, and the dotted orange lines indicate smoothed values. The bottom row presents evaluation metrics, including recall, precision, mean Average Precision at IoU threshold 0.50 (mAP50) and across multiple IoU thresholds from 0.50 to 0.95 (mAP50–95). The metrics show improving performance over training epochs, with recall, precision, and mAP metrics approaching high values. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

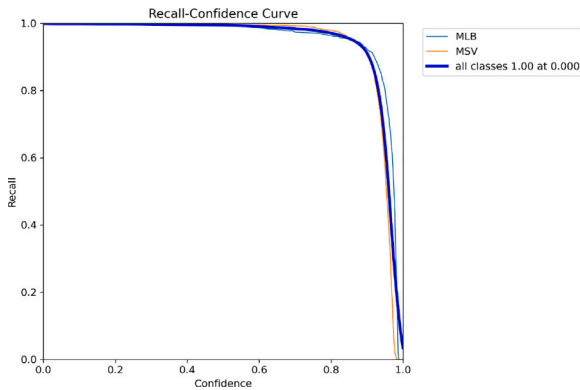


Fig. 14. YOLO v10 recall-confidence curve; the model detection achieved 1.00 for all classes. The curves demonstrate the model's ability to balance recall and confidence effectively.

generalization capabilities. Fig. 20 provides visual examples of maize rust, illustrating the characteristic features initially misclassified as MLN.

The comparison between actual and predicted disease counts and confidence intervals are shown in Fig. 21. This indicated high model reliability for most classes, except MLN, which exhibited broader confidence intervals due to limited data. Based on the field test results, the best classification model was retrained to include maize rust as a distinct class. The retrained model demonstrated improved accuracy in classifying maize rust, as shown in the confusion matrix in Fig. 22. Of the 25 images we had for the maize rust test class, only two were classified wrongly as NL or MLB. This showcases that the model proposed in this remaster works on maize rust and the considered classes before the field test.

The field tests validated the MaizeNet application's potential for real-time maize disease diagnosis. The expert input feature enhanced diagnostic accuracy and provided valuable data for model refinement. The challenges encountered with MLN and maize rust highlight the importance of diverse and representative datasets. Retraining the model with maize rust significantly improved its performance, demonstrating the system's adaptability.

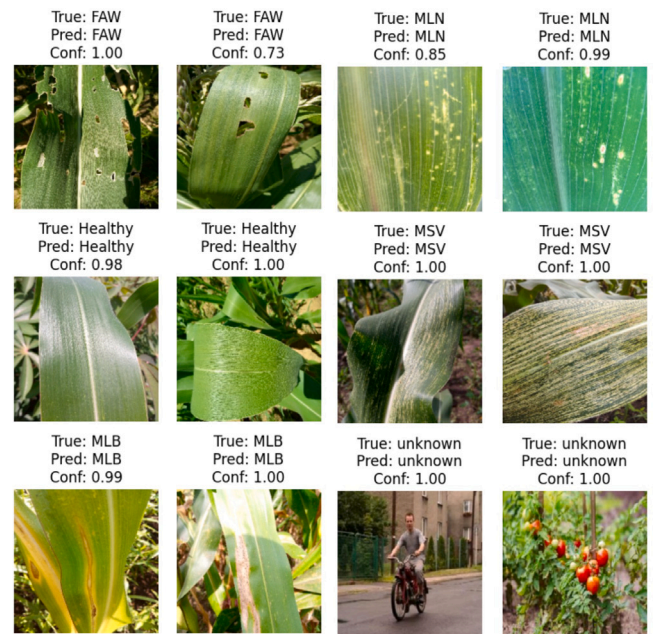


Fig. 15. Model inference results on images, showcasing examples of correctly classified instances. Each sub-image displays the ground truth label, the model's predicted label, and the confidence score. The last row includes a sample of unknown categories, with the model accurately identifying them as 'unknown' with high confidence, demonstrating its ability to handle out-of-distribution samples.

4. Discussion

This study demonstrates the effectiveness of using machine learning models to diagnose maize leaf diseases in sub-Saharan Africa. Integrating deep learning and traditional methodologies provided a comprehensive assessment of accurate disease classification and detection. Applying transformer-based architectures, notably Vision Transformers, and state-of-the-art deep pre-trained models, such as InceptionResNetV2, via transfer learning, demonstrated significant potential in capturing complex feature representations critical for distinguishing maize leaf diseases. Furthermore, developing a custom CNN tailored for maize classification showcased the ability to create specialized models capable of learning intricate disease-specific features. These findings align with recent studies highlighting the effectiveness of deep learning in agricultural disease detection (Ferentinos, 2018; Mohanty et al., 2016).

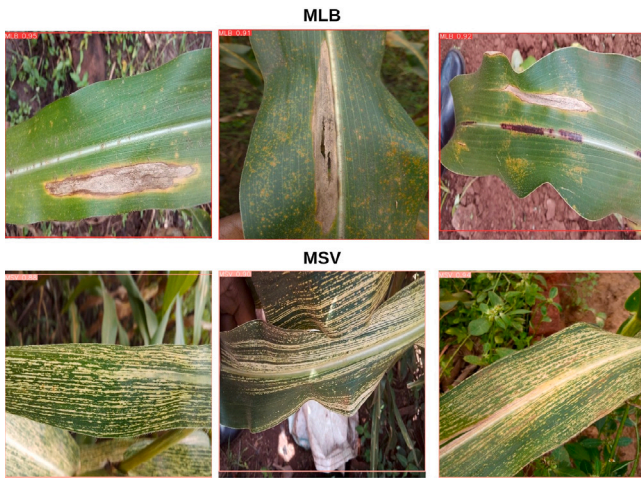


Fig. 16. Detection inference results for MLB and MSV. The images show the model accurately identified lesions with confidence scores displayed in the red bounding boxes. The top row illustrates examples of MLB, and the bottom row shows examples of MSV.

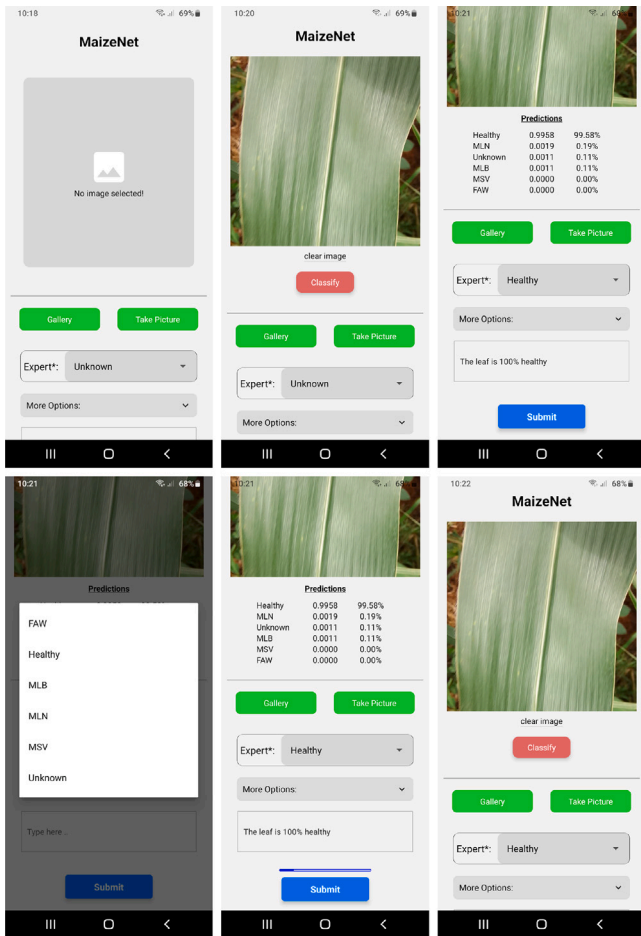


Fig. 17. Interface walkthrough of the MaizeNet application for disease diagnosis in maize crops: (a) Image selection, (b) Classification, (c) Prediction results, (d) Expert input, and (e) Submission.

The ensemble model approach significantly improved classification accuracy by leveraging the strengths of multiple models. This strategy effectively mitigated overfitting, a common challenge in deep learning, and enhanced the model's generalization capabilities across diverse

datasets. Ensemble methods have been widely recognized for their robustness and ability to improve predictive performance in various domains (Dietterich, 2000).

In object detection, integrating transformer components in the YOLOv10 model marked a significant advancement. Its ability to handle complex scenes, accurately detect overlapping objects, and perform real-time disease detection with high precision makes it a valuable tool for agricultural diagnostics. The model had an mAP of 0.995 across different disease classes, highlighting YOLOv10's ability in disease detection. Similarly, the Detection Transformer model demonstrated its effectiveness in accurately localizing and identifying disease symptoms by capturing spatial relationships and contextual information within images (Carion et al., 2020).

This work also contributes to the integration of Explainable AI techniques, specifically Grad-CAM and LIME, to provide insights into the model's decision-making process. Grad-CAM visualizations for the MobileNetV2 model highlighted the specific image regions that influenced the predictions, aligning with expert knowledge of disease symptoms. LIME complemented Grad-CAM by providing pixel-level explanations, demonstrating the model's reliability and fostering trust among end-users. The ability of LIME to identify that the model does not utilize any disease-specific features when classifying the 'unknown' class is crucial for novel disease identification. XAI techniques are increasingly recognized as essential for enhancing transparency and trust in AI applications, particularly in critical domains like agriculture. However, while explainability techniques were employed during the model evaluation phase, they were not integrated into the deployed application. These tools were primarily used to interpret and verify the model's predictions, consistently highlighting disease-relevant regions and confirming the model's reliability and transparency. Given this strong alignment and the high performance of the model, it was determined that incorporating XAI into the field application would add unnecessary complexity without providing significant practical value. Additionally, omitting XAI components ensured faster inference and a more efficient user experience, which was a critical consideration for deployment in resource-limited, real-time agricultural environments.

The field test results revealed the model's high accuracy in classifying well-represented diseases, including FAW, MSV, MLB, and healthy individuals. Although the model performed well overall, its relatively poor performance on rare classes, such as MLN, was primarily due to the limited representation of these classes in the training data. The misclassification between maize rust and MLN highlights the challenges of visually distinguishing between similar diseases when few examples are available. To address this issue, future work will focus on collecting more diverse and balanced datasets, particularly for underrepresented disease categories. Additionally, we plan to explore synthetic data augmentation techniques, such as advanced image transformations, as well as active learning strategies that prioritize uncertain or misclassified samples. These efforts aim to enhance the model's robustness and generalizability. The inclusion of non-target diseases in the 'Unknown' class testing provided valuable insights into the model's ability to handle out-of-scope inputs. Furthermore, retraining the model with maize rust data improved its performance, demonstrating the potential for iterative model refinement. Expert feedback and annotations during field testing emphasized the importance of expert-in-the-loop approaches to enhance model adaptability and expand its applicability.

5. Conclusion

This study utilizes diverse and real-world maize leaf datasets from various regions to develop effective disease detection models. It examines and compares classical methods, deep learning techniques, ensemble approaches, and interpretable AI methods while incorporating optimized detection modules. Additionally, the research demonstrates successful deployment on edge devices, along with field testing for



Fig. 18. Field testing of the MaizeNet application on edge devices.

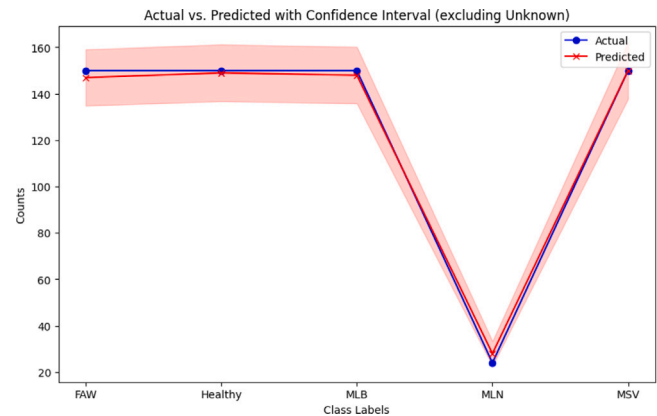


Fig. 19. Confusion matrix showing the MaizeNet model performance.

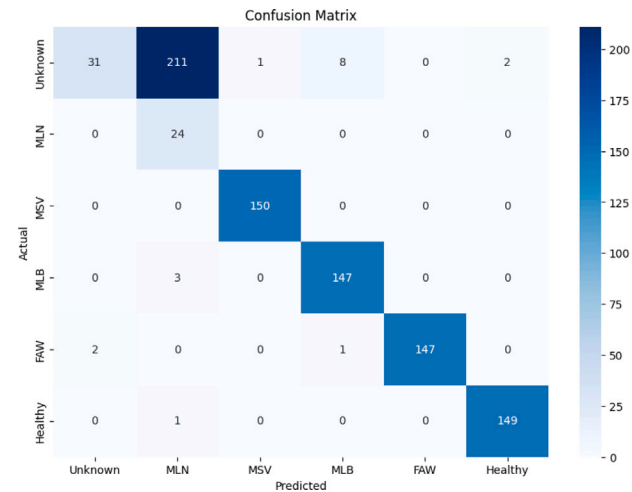


Fig. 21. Confusion matrix showing MaizeNet model performance with expert-labeled data.



Fig. 20. Example images of maize leaves affected by maize rust disease.

practical, in-situ evaluation. The results indicate that models like MobileNetV2, InceptionResNetV2, and ViT provide high accuracy and robustness, making them suitable for deployment in practical agricultural scenarios. Specifically, MobileNetV2 not only offered the best performance (97% accuracy) but also maintained a lightweight architecture ideal for mobile deployment, making it the most suitable choice for real-time, resource-constrained environments. The ensemble model approach further enhances performance by combining the strengths of different architectures, providing a comprehensive solution for detecting maize disease.

The success of YOLOv10 and DETR models in object detection highlights the importance of incorporating advanced features like transformer components to handle complex real-world scenarios. These models' ability to provide accurate and real-time detection makes them valuable tools for farmers, enabling early intervention and reducing the impact of maize diseases on crop yields.

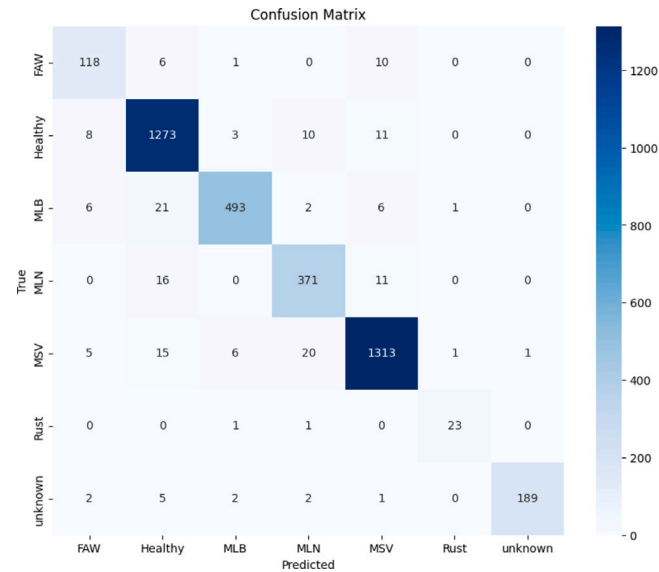


Fig. 22. Confusion matrix showing the performance of the retrained MobileNet model.

The MaizeNet model effectively classifies major maize diseases, achieving accurate results for primary disease categories and providing a practical tool for real-time field diagnostics. Despite high accuracy for common diseases, the model's performance with rare and visually similar diseases indicates areas for future improvement. By expanding the dataset to include additional rare and non-target diseases and leveraging expert input, MaizeNet can evolve to become a more comprehensive and reliable tool for supporting maize farmers and reducing crop losses due to misdiagnosis.

Future research should focus on further improving model efficiency, reducing computational requirements, and exploring the integration of more diverse datasets to enhance model generalization. Developing lightweight models optimized for mobile devices will be crucial for expanding access to these advanced AI tools, particularly in resource-constrained settings. By continuing to advance these technologies, we can support sustainable agriculture and improve food security worldwide.

CRedit authorship contribution statement

Joyce Nakatumba-Nabende: Conceptualization, Methodology, Writing – original draft, Supervision, Writing – review. **Sudi Murindanyi:** Data curation, Writing – original draft, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared the link to the datasets in the paper.

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